Layout-Anchored Prioritizing Continual Learning for Continuous Building Footprint Extraction From High-Resolution Remote Sensing Imagery

Dingyuan Chen, Zhaohui Song, *Senior Member, IEEE*, Ailong Ma, *Senior Member, IEEE*, and Yanfei Zhong, *Senior Member, IEEE*

***Abstract*— Continuous building footprint extraction requires learning new building patterns from remote sensing imagery without forgetting old knowledge. It is inherently challenging due to the spatial layout heterogeneity, which leads to the problem of knowledge forgetting in two aspects: complex background could have distinct patterns (background diversity) and buildings could have similar patterns to the background (foreground–background similarity). To solve the issues, we propose a domain-incremental continual learning algorithm named layout-anchored prioritizing learning network (LAPNet), including a latent layout anchoring module and layout-aware prioritizing learning module. The latent layout anchoring aggregates background information into latent layout features and employs a herding strategy to select representative layout anchors iteratively. This module maintains a memory buffer to narrow the background differences by dynamically discarding unrepresentative experiences and storing layout-anchored experiences. Furthermore, layout-aware prioritizing learning uses these experiences to identify and emphasize the most valuable knowledge for maximizing interclass distance. This module leverages the layout variance metric to measure interclass discrepancies and employs prioritizing learning to reweight the optimization function based on this layout prior. We established a Global-CL dataset to validate the proposed LAPNet framework, containing six study areas across four continents with different remote sensing sensors. Experiments showed that LAPNet achieves state-of-the-art performance in continuous building footprint extraction by effectively correlating knowledge across various domains. The code is available at: https://github.com/Dingyuan-Chen/LAPNet.**

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Dingyuan Chen is with the Space Information Research Institute and Zhejiang Key Laboratory of Space Information Sensing and Transmission, Hangzhou Dianzi University, Hangzhou 310018, China, and also with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University, Wuhan 430079, China (e-mail: chendingyuan@hdu.edu.cn).

Zhaohui Song is with the Space Information Research Institute and Zhejiang Key Laboratory of Space Information Sensing and Transmission, Hangzhou Dianzi University, Hangzhou 310018, China (e-mail: songzh@hdu.edu.cn).

Ailong Ma and Yanfei Zhong are with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University, Wuhan 430079, China (e-mail: maailong007@whu.edu.cn; zhongyanfei@whu.edu.cn).

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***Index Terms*— Continual learning, continuous building footprint extraction, layout-anchored prioritizing learning, remote sensing, spatial layout heterogeneity.**

I. Introduction

BUILDING footprints provide fundamental information for urbanization analysis and humanitarian aid [1], [2], [3], [4], [5]. Organizations like Microsoft have released open-source building footprints to provide a comprehensive view of the built environment [6], [7], [8], [9], [10]. However, due to the demands of dramatic urban expansion and disaster emergency [11], [12], [13], [14], developing a dynamic model that can effectively handle continuously available remote sensing imageries is essential for continuous building footprint extraction. Its critical step is knowledge accumulation, efficiently learning from current, and historic building patterns without forgetting previously acquired knowledge.

Recently, domain-incremental continual learning methods have achieved great success in enabling knowledge accumulation by allowing a model to learn from multiple domains while retaining previously acquired information sequentially, which benefits the development of continuous building footprint extraction [15], [16], [17], [18]. Domain-incremental continual learning methods accumulate knowledge by retaining important parameters, utilizing knowledge distillation, and replaying representative experiences, which can be roughly divided into parameter regularization methods, functional regularization methods, and replay methods. Parameter regularization methods constrain the changes of parameters necessary for past domains, such as EWC [19] and SI [20]. In complex remote sensing scenarios, DILRS [21] uses domain residual adapter modules to mitigate label space shifts, while UNISA [22] employs unsupervised flat-wide learning to tackle data scarcity in continual learning. Functional regularization methods solve the knowledge-forgetting problem by constraining the input-output mapping function, such as LwF [23]. A continual learning-guided training framework [24] incorporates knowledge distillation to apply holistic constraints and prevent large weight variations, while incremental learning for semantic segmentation [25] uses knowledge distillation to balance output discrepancies between the memory and the

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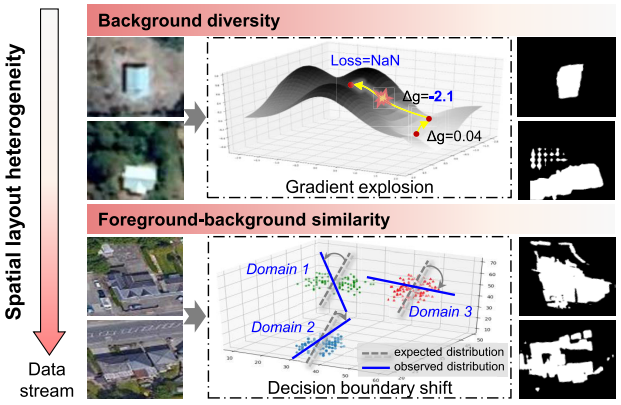


Fig. 1. Spatial layout heterogeneity contributes to knowledge forgetting in two aspects: 1) background diversity causes a gradient explosion and 2) foreground–background similarity results in a decision boundary shift.

updated network. Replay methods explicitly store the past experiences into a memory or a generative model and rehearse them back with the new input samples, such as A-GEM [26] and DGR [27]. CL-CaGAN [28] learns representative spectral characteristics to estimate the background distribution and provide spectral prior information, while IDHT [29] develops an incremental spectral detector to align latent feature spaces effectively.

However, continuous building footprint extraction suffers from spatial layout heterogeneity, leading to knowledge forgetting in two aspects: background diversity and foreground–background similarity, as shown in Fig. 1. The background diversity stems from diverse topography and landscape, including hills and plains at different elevations and varied environments such as bare land and vegetation. The background diversity could cause a gradient explosion and affect the stability of knowledge accumulation. Formally, the discrepancy of feature representations between the current and past domains results in the contradiction of the gradient direction. Therefore, aggregating representative features is crucial for stable knowledge accumulation. On the other hand, the foreground–background similarity is because the foreground and background exhibit significant spatial randomness, resulting in a higher incidence of similar architectural materials and sensor radiation prone to interclass confusion. The foreground–background similarity may lead to a decision boundary shift even with a minor disturbance and hinders knowledge accumulation in long-term memory. Specifically, the model parameters gradually shift and align more with the characteristics of the new domains, potentially neglecting the previously consolidated knowledge. This divergence between the expected and observed distributions highlights the challenge of maintaining model stability. Therefore, an adaptive learning strategy should be developed to maximize the interclass distance and continuously achieve high-accuracy performance.

The two key steps to address spatial layout heterogeneity are the **storage** and **replay** of representative experiences. Storage strategy selects and maintains representative experiences to mitigate the issue of intraclass diversity [30], [31], [32], [33]. iCaRL [31] estimates representative features by calculating the average feature vectors over all exemplars to select representative experiences optimally. GCR [30] and GSS-Greedy [32] maintain a subset that closely approximates the gradient of all the data seen so far to identify representative experiences. In dynamic memory [33], a high-level style metric is used to select representative images for dynamic memory. However, these methods treat the foreground and background equally without considering the layout prior of complex background. This results in biased estimates of representative experiences since the layout diversity occurs mainly in the background. Replay strategy aims at aggregating experiences with significant interclass discrepancies to alleviate the issue of interclass similarity [16], [34], [35], [36]. In CLOPS [16], uncertainty-based acquisition functions are used to discard instances close to the decision boundary, which helps to maximize the interclass distance. PER [36] selects the experience that maximally reduces the global loss in its current domain to consolidate knowledge. Although these methods provide a global representation of interclass discrepancies, they are required to capture subtle layout clues that distinguish the foreground from the background, resulting in inefficient learning during knowledge accumulation. Therefore, existing storage and replay strategies face the issue of knowledge forgetting due to background diversity and the similarity between foreground and background.

In this article, we introduce a domain-incremental continual learning algorithm named layout-anchored prioritizing learning network (LAPNet) to solve the problem of knowledge forgetting caused by spatial layout heterogeneity. The main contributions are as follows.

1) A LAPNet framework is proposed to deal with spatial layout heterogeneity for continuous building footprint extraction, including latent layout anchoring and layout-aware prioritizing learning modules. To reduce background variability, latent layout anchoring incorporates layout priors of complex backgrounds to identify representative experiences, ensuring stable knowledge accumulation. To maximize the interclass distance, the layout-aware prioritizing learning measures the layout variance metric to capture subtle layout clues for prioritizing learning.
2) A latent layout anchoring module is constructed to eliminate the background diversity. This module anchors experiences of different domains to the same latent space by learning latent layout features of complex backgrounds. Specifically, the latent layout anchoring explicitly aggregates the background into latent layout features and employs a herding strategy to iteratively identify layout anchors, referring to feature vectors located around feature centers. This module maintains a memory buffer to narrow the background gap by dynamically discarding unrepresentative experiences and storing layout-anchored experiences.
3) A layout-aware prioritizing learning module addresses the challenges arising from foreground–background similarity. It follows the idea of adaptively identifying the most valuable knowledge to maximize the

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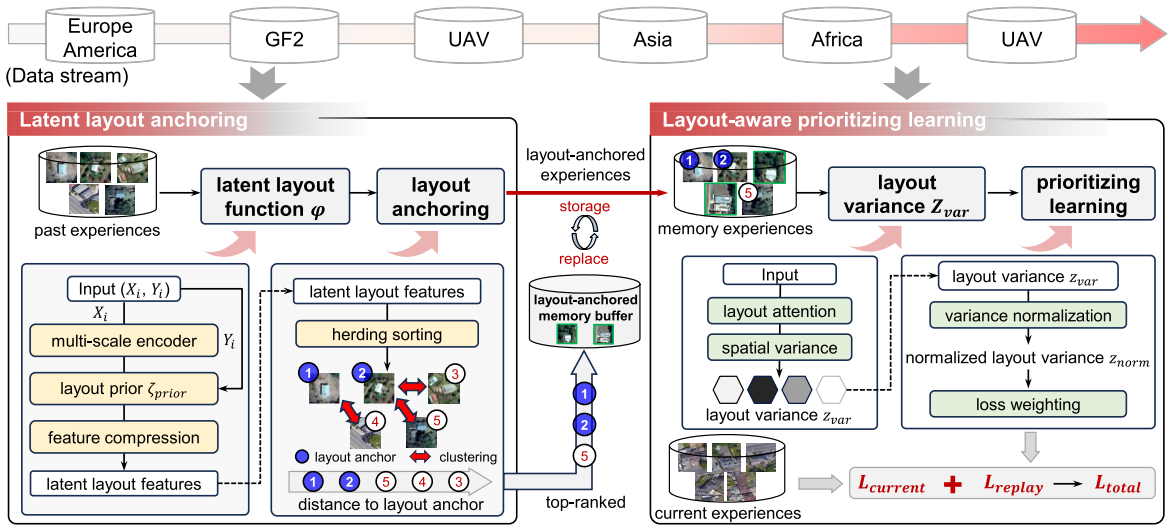


Fig. 2. Overview of the proposed LAPNet framework. It consists of a latent layout anchoring module to identify representative experiences for stable knowledge accumulation and a layout-aware prioritizing learning module to identify the most valuable knowledge for efficient model learning adaptively.

interclass distance for efficient model learning. Indeed, the layout-aware prioritizing learning leverages layout variance, a metric that captures subtle layout clues to measure interclass discrepancies. It estimates fine-grained layout relationships between foreground and background by assessing the relative differences of latent layout features. Furthermore, based on this layout prior, this module employs prioritizing learning to reweight the optimization function at the image level within each batch.

We organize the rest of this article as follows. Section II proposes the LAPNet framework and describes two submodules in detail. Section III introduces the experimental settings and experimental results. In Section IV, we discuss the ablation experimental results of each module. We conclude this research and reveal future potential works in Section V.

II. Proposed LAPNet Framework

In this section, the proposed LAPNet framework is introduced in detail. It consists of a latent layout anchoring module and a layout-aware prioritizing learning module. As shown in Fig. 2, a sequence of data streams is fed into the proposed LAPNet framework for continuous building footprint extraction. First, the data streams are sequentially entered into the latent layout anchoring module to identify and store the layout-anchored experiences, which helps to narrow the background gap during knowledge accumulation. Furthermore, the layout-anchored experiences are then fed into the layout-aware prioritizing learning module to capture subtle layout clues based on the layout variance prior, thereby maximizing the interclass distance for efficient learning.

A. Latent Layout Anchoring

The latent layout anchoring module aims to identify representative experiences for stable knowledge accumulation. As shown in Fig. 3, this module explicitly relates representative experiences to the layout anchors, which introduces the

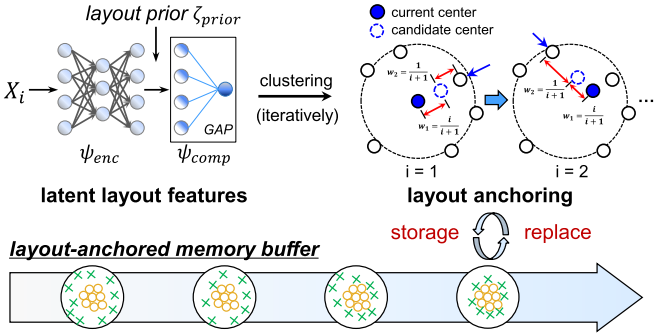


Fig. 3. Architecture of the latent layout anchoring module. It encodes inputs into the latent layout features, explicitly relates representative experiences to the layout anchors, and dynamically updates layout-anchored memory buffer for stable knowledge accumulation.

layout priors of complex backgrounds into latent feature representation for identifying layout-anchored experiences. These experiences are iteratively stored within a layout-anchored memory buffer to replace unrepresentative experiences dynamically. Next, the layout anchor and the layout-anchored memory buffer are described in detail.

1) Layout Anchor: Layout anchor is defined as feature vectors associated with representative experiences, designed to anchor experiences from diverse domains within the same latent space. Initially, a latent layout function φ is devised to encode input images into latent feature representations. Similar latent features are clustered using a herding strategy, with feature centers to identify layout anchors.

The latent layout function φ aims at explicitly aggregating the background into latent layout features. Specifically, an image X_i equipped with a label Y_i is fed into a layout feature encoder, followed by feature compression, to generate the latent layout features. This process can be described as follows:

$$\varphi = \psi_{\text{comp}}(\zeta_{\text{prior}}(\psi_{\text{enc}}(X_i), Y_i)) \tag{1}$$

where ψ_{enc}, ψ_{comp}, and ζ_{prior} represent the layout feature encoder, feature compression, and layout prior function that

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explicitly models complex backgrounds, respectively. The layout feature encoder is designed to perceive multiscale objects arising from domain gaps due to terrain variance and sensor resolution differences. It integrates multiscale layout representations using a ResNet backbone equipped with the atrous spatial pyramid pooling (ASPP) [38]. The ASPP employs atrous convolutions at 12, 24, and 36 rates to capture global layout semantics, reflecting the spatial arrangement of building footprints and local details essential for precisely defining building boundaries.

After obtaining general layout information, a layout prior function is proposed to inject background priors into feature aggregation explicitly. It uses the label Y_i to provide pixel-level location of complex backgrounds. The layout prior function can be mathematically defined as

$$\zeta_{\text{prior}}(\psi_{\text{enc}}(X_i), Y_i) = \psi_{\text{enc}}(X_i)[m, n] * \mathcal{P}[c, m, n] \quad (2)$$

where $\mathcal{P}$ represents the layout prior indexing mechanism. c, m, and n represent the cth channel, mth column, and nth row, respectively.

By setting c to 0, the background prior is indexed for layout feature aggregation—such a background prior aims at capturing representative spatial layout heterogeneity for layout anchoring. Furthermore, feature compression is adopted to identify discriminative features and discard redundant features. Specifically, a global average pooling (GAP) [39] is used for global layout feature compression, which follows the idea of global channel attention to activate important channel information and discard less informative channels effectively.

Furthermore, layout anchors clustered around feature centers are located using a herding sorting strategy [37]. Initially, a centroid that represents the average position of the feature space is selected as the initial anchor point. Then, all the candidate points are ranked based on their similarity to the centroid, where the similarity is measured based on the Euclidean distance. The top-ranked points are selected and treated as layout anchors. The layout anchors selected in the current iteration are used to update the centroid. The updated centroid represents the refined positions in the feature space. The iterations continue until a certain number of layout anchors are met, and the selected anchors are considered the final set of layout anchors.

2) Layout-Anchored Memory Buffer: A layout-anchored memory buffer is maintained for knowledge storage to identify representative experiences based on layout anchors. As described in Algorithm 1, given an exemplar set A, we aim to select the most representative m experiences to store in the memory buffer $\mathcal{B}$.

There are two storage strategies in layout-anchored memory buffer: basic and layout-anchored. If the memory buffer is sufficient for storing all the exemplars, i.e., $|\mathcal{B}| \geq n$, the basic storage is used to transfer the exemplar set A to a memory buffer $\mathcal{B}$. Otherwise, the layout-anchored storage is used to keep the most representative m experiences by iteratively locating layout anchors that are clustered around feature centers. The m experiences are then fed into the layout-anchored memory buffer to best approximate the overall data distribution across all training examples.

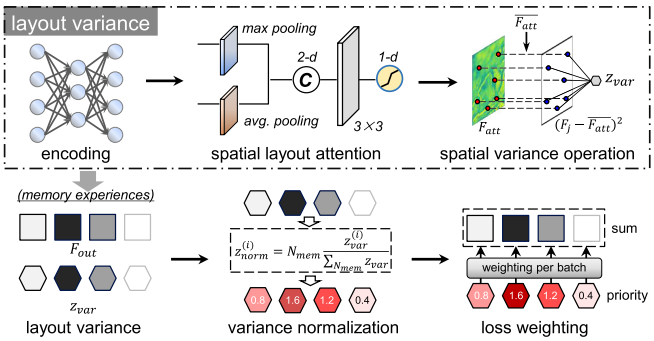


Fig. 4. Architecture of the layout-aware prioritizing learning module. It generates a layout variance metric to measure fine-grained foreground–background discrepancies, and uses a prioritizing learning mechanism to adaptively identify the most valuable knowledge for dealing with the interclass similarity effectively.

When the memory buffer is full, unrepresentative experiences should be replaced by the current layout-anchored experiences. Formally, our goal is to reduce the number of exemplars from m' to m, one discards the experiences $p_{m+1}, \ldots, p_{m'}$, keeping only the experiences $p_1, \ldots, p_m$. This procedure retains only the most representative m experiences for storage while discarding those away from the feature centers.

Algorithm 1 Layout-Anchored Memory Buffer

Input: exemplar set $A = \{(x_1, y_1), \ldots, (x_n, y_n)\}$
Input: memory buffer $\mathcal{B}$
Input: m target number of exemplars
Require: latent layout function $\varphi : \mathcal{X} \to \mathbb{R}^d$

```
if |B| ≥ n then
    B ← A                                   ▷ basic storage
else
    μ ← (1/n) Σ_{(x,y)∈A} φ(x, y = 0)
    for j = 1, ..., m do
        p_j ← arg min_{(x,y)∈A} ‖μ − (1/j)[φ(x, y = 0) + Σ_{k=1}^{j−1} φ(p_k)]‖
    end for
    B ← (p_1, ..., p_m)                     ▷ layout-anchored storage
end if
```

B. Layout-Aware Prioritizing Learning

The proposed LAPNet integrates a layout-aware prioritization learning module that adaptively identifies the most valuable knowledge for efficient model learning, aiming to maximize interclass distance. As shown in Fig. 4, a layout variance prior to capturing subtle layout clues is introduced to measure interclass discrepancies. Based on this layout prior, a prioritizing learning mechanism is implemented to reweight the optimization function, effectively addressing interclass similarity.

1) Layout Variance Prior: The layout variance metric measures fine-grained interclass divergencies by introducing the intrinsic relationship of latent layout features. It takes a feature map $F_i \in \mathbb{R}^{C \times W \times H}$ from the ResNet backbone to generate a layout attention map F_{att} with fine-grained layout information.

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The F_{att} is then fed into the spatial variance operation to generate the layout variance metric Z_{var}.

To obtain subtle layout clues, we introduce the spatial layout attention mechanism to capture discriminative layout features. It aggregates activated features by incorporating a max pooling and an average pooling, which are followed by a channel concatenation function, a $k \times k$ ($k = 3$) convolutional layer, and a sigmoid function to fuse fine-grained layout information. The spatial layout attention is formulated as follows:

$$F_{\text{att}} = \sigma\left(\text{cat}(\text{Max}P(F_i), \text{Avg}P(F_i)) * K^{(1)}\right) \tag{3}$$

where Max$P$, Avg$P$, cat, $K^{(1)}$, and σ represent the max pooling, average pooling, channel concatenation function, convolutional layer, and sigmoid function, respectively.

After spatial layout attention, we incorporate F_{att} into the input feature map F_i to enhance layout semantics for improving the performance of building footprint extraction. This strategy can be formulated as follows:

$$F_i' = F_i * F_{\text{att}}. \tag{4}$$

The spatial variance operation inputs F_{att} to generate the layout variance metric Z_{var}. Specifically, this operation estimates the gap between the features of a certain location and the average features and aggregates layout details to capture intrinsic interclass relationships. The average features $\overline{F_{\text{att}}}$ can be defined as

$$\overline{F_{\text{att}}} = \frac{\sum_{i=1}^{n} F_{\text{att}}^{(i)}}{n} \tag{5}$$

where n is the number of feature points in F_{att}.

The layout variance metric Z_{var} estimates fine-grained layout relationships between foreground and background by assessing the relative differences of latent layout features. The insight is that the relative difference remains stable even if the decision boundary is shifted during knowledge accumulation. By accumulating the layout semantics of each point, subtle layout clues can be used to alleviate the interclass similarity. Mathematically, Z_{var} can be formulated as

$$Z_{\text{var}} = \frac{\sum_{i=1}^{n}\left(F_{\text{att}}^{(i)} - \overline{F_{\text{att}}}\right)^2}{n-1}. \tag{6}$$

2) Prioritizing Learning Based on Layout Variance: After obtaining the layout variance prior, prioritizing learning reweights the optimization function to accelerate the backpropagation of important knowledge. Given a predicted output F_{out} and a label Y, the layout variance Z_{var} is adopted for loss weighting to maximize the interclass distance. The Z_{var} is first normalized to Z_{norm} and fed into the optimization function within each batch for layout-aware prioritizing learning.

The enhanced features F_i' are decoded into a predicted output F_{out} by constructing a decoder φ_{dec}. The generation of F_{out} is formulated as follows:

$$F_{\text{out}} = \varphi_{\text{dec}}\left(F_i'\right) \tag{7}$$

$$\varphi_{\text{dec}}\left(F_i'\right) = \text{up}\left(\text{ASPP}\left(F_i'\right) * K^{(2)}\right) \tag{8}$$

where the φ_{dec} includes an ASPP module ASPP, a convolutional layer $K^{(2)}$ with a stride of 1, and an upsampling layer up with a scale of 8.

The variance normalization converts Z_{var} to the normalized layout variance Z_{norm}, ensuring the stability of the optimization. Indeed, the sum of weights is normalized to the batch size N_{BS}, which can be expressed as

$$Z_{\text{norm}}^{(i)} = N_{\text{BS}} \frac{Z_{\text{var}}^{(i)}}{\sum_{j=1}^{N_{\text{BS}}} Z_{\text{var}}^{(j)}}. \tag{9}$$

The total loss function incudes $\mathcal{L}_{\text{current}}$ for the current experiences and $\mathcal{L}_{\text{replay}}$ for the memory experiences. It aims to learn new building patterns while preserving old knowledge. The total loss is expressed as

$$\mathcal{L}_{\text{total}} = \frac{1}{N_{\text{domains}}}\mathcal{L}_{\text{current}} + \left(1 - \frac{1}{N_{\text{domains}}}\right)\mathcal{L}_{\text{replay}} \tag{10}$$

where N_{domains} represents the number of domains seen so far, a dynamic parameter that balances the two loss terms, as the N_{domains} increases, more attention is paid to previous domains to avoid forgetting knowledge.

Both the $\mathcal{L}_{\text{current}}$ and $\mathcal{L}_{\text{replay}}$ are made up of binary cross-entropy loss [40] and dice loss [41]. Differing from the original version, the prioritizing learning mechanism is injected into $\mathcal{L}_{\text{bce_prior}}$ and $\mathcal{L}_{\text{dice_prior}}$ in the $\mathcal{L}_{\text{replay}}$. Mathematically, $\mathcal{L}_{\text{replay}}$ can be represented as

$$\mathcal{L}_{\text{replay}} = \mathcal{L}_{\text{bce_prior}} + \mathcal{L}_{\text{dice_prior}}. \tag{11}$$

The layout-aware prioritizing learning is conducted based on the layout variance prior. Specifically, Z_{norm} is used as a weighted term to dynamically rank the importance of experiences. This process identifies the most valuable knowledge to maximize interclass distance, thereby enhancing convergence efficiency. The loss functions $\mathcal{L}_{\text{bce_prior}}$ and $\mathcal{L}_{\text{dice_prior}}$ are defined as follows:

$$\mathcal{L}_{\text{bce_prior}} = -\frac{1}{N}\sum_{i}^{N} Z_{\text{norm}}^{(i)} \mathcal{L}_{\text{bce_prior}}^{(i)} \tag{12}$$

$$\mathcal{L}_{\text{bce_prior}}^{(i)} = \left(1 - F_{\text{out}}^{(i)}\right)\log(1 - Y_i) + F_{\text{out}}^{(i)} \log Y_i \tag{13}$$

$$\mathcal{L}_{\text{dice_prior}} = 1 - 2\mathcal{Z}\frac{|A \cap B|}{|A| + |B|} \tag{14}$$

where $F_{\text{out}}^{(i)}$ represents the ith predicted output, Y_i represents the ith true value, A represents a collection of the predicted values, B represents the true values, and $\mathcal{Z}$ represents a collection of the normalized layout variance Z_{norm}.

III. Experiments

A. Datasets and Preprocessing

To evaluate the performance of continuous building footprint extraction methods, we conducted experiments using a global building footprint continual learning (Global-CL) dataset, and the main characteristics of which are summarized in Table I. As shown in Fig. 5, the test subset of the Global-CL dataset contains six study areas across four continents with different remote sensing sensors, which divides the spatial layout heterogeneity into cross-sensor and cross-region settings. The cross-sensor tasks involve data from Google Earth (GE), GF-2, and UAV-borne sensors, while the cross-region tasks cover areas in Europe, America, Asia, and Africa. The images were

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TABLE I

GLOBAL-CL DATASET COMPRISES CROSS-SENSOR TASKS INTEGRATING GE, GF-2, AND UAV-BORNE SENSORS, AS WELL AS CROSS-REGION TASKS SPANNING REGIONS ACROSS EUROPE, AMERICA, ASIA, AND AFRICA

Task	Cross-X	Sensor	Region	Train	Test	Collection year (train | test)
1	-	GE(0.5m)	Europe and America (Las Vegas, Paris)	4999 (labeled)	1013 (Boston, Birmingham)	2021 | 2019
2	Sensor	GF-2(0.8m)	Wuhan	5448 (unlabeled)	273	2018;2022 | 2015;2016;2018
3	Sensor	UAV(0.3m)	Nanjing	7000 (unlabeled)	3338	2020 | 2020
4	Region	GE(0.5m)	Asia (Tokyo)	7940 (unlabeled)	289	2021 | 2021
5	Region	GE(0.5m)	Africa (Nairobi, Kigali)	4764 (unlabeled)	4480	2021 | 2021
6	Sensor	UAV(0.09m)	Tallinn	7200 (unlabeled)	386	2021 | 2021

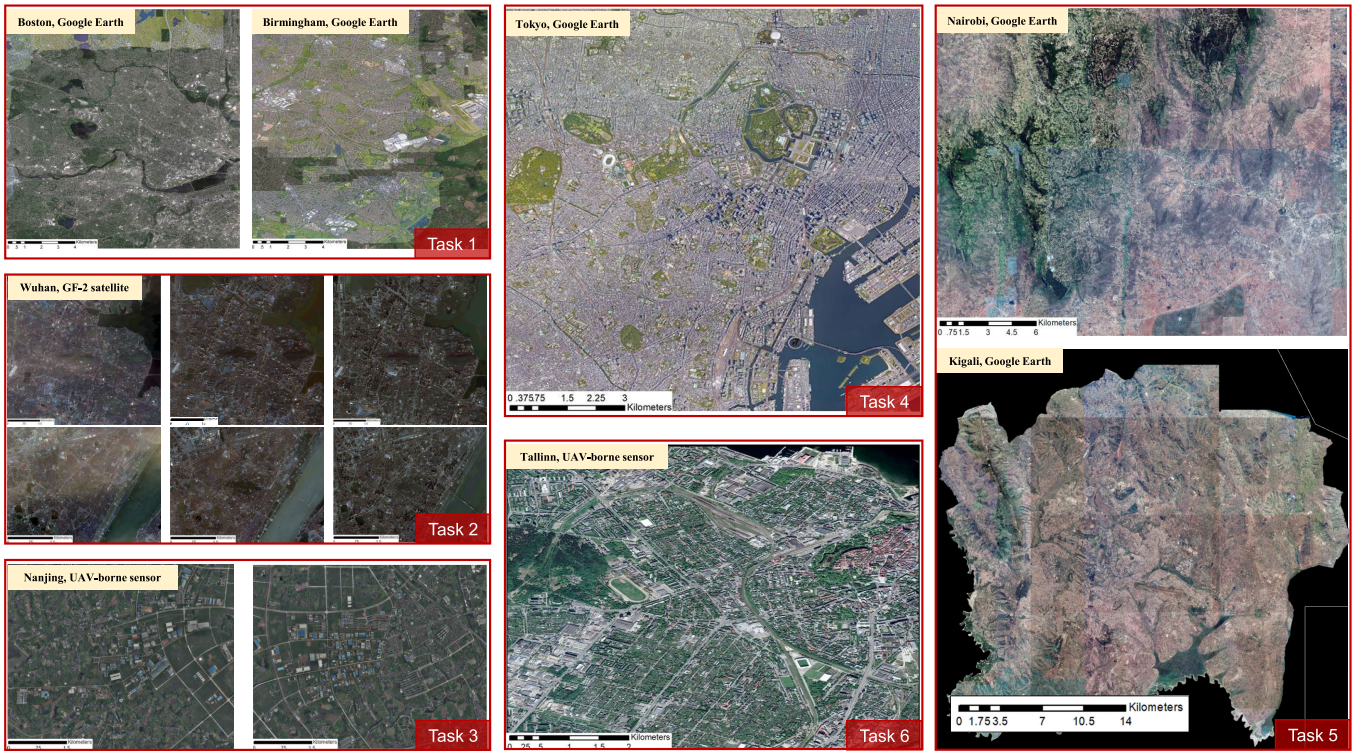


Fig. 5. Global-CL dataset visualization. Task 1 involves GE imageries from Europe (Birmingham) and American (Boston) regions, Task 2 represents GF-2 data collected in Wuhan, Task 3 represents UAV-borne sensor imageries from Nanjing, Task 4 involves GE imageries from Asia (Tokyo), Task 5 involves GE imageries from Africa (Nairobi and Kigali), and Task 6 represents UAV-borne sensor imageries from Tallinn.

cropped to tiles of 1024 × 1024 pixels, preserving the red, green, and blue channels.

To simulate real-world scenarios, we limited our data annotations to the source domain (Task 1), leaving the remaining domains (Tasks 2–6) unlabeled. To facilitate domain-incremental continual learning, we employed GUMONet [42], a semi-supervised model explicitly tailored for global building footprint extraction, to generate pseudo-labels for Tasks 2–6 by learning from the labeled data in Task 1. Each task (denoted as T_i) was associated with an individual model M_i, where the training strategy of the model M_i is denoted as "individual training." The tasks were arranged in ascending order according to their test set accuracy following the "individual training" strategy. Additionally, we conducted simple baseline experiments, including fine-tuning a model incrementally across domains (referred to as the "naïve" approach) and training a unified model using all domain training data (referred to as the "unified training" framework). Each task is subsequently evaluated using dedicated test sets.

B. Experimental Settings

The experiments were conducted on a single Tesla P100 graphics processing unit (GPU), and all the methods were implemented using PyTorch [43]. We employed DeepLab v3 [38] as our primary network. The backbone was initialized with the released ResNet-34 model [44] pretrained on ImageNet [45]. Stochastic gradient descent with a learning rate of 0.03 and momentum of 0.9 was used for the optimization. The experiments were conducted for 150 epochs, with 25 epochs per task and a batch size 16.

To assess the performance of the proposed LAPNet framework, we compared it with 1) the parameter regularization methods of EWC [19] and SI [20]; 2) the functional

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regularization methods of LwF [23], ILT [46], BiC [47], MiB [48], PLOP [49], and LGKD [50]; and 3) the replay methods of ER [51], iCaRL [31], GSS-Greedy [32], A-GEM [26], and DGR [27]. For parameter regularization methods, the regularization hyperparameters λ were set to 0.4 for EWC and 0.0001 for SI. For functional regularization methods, the temperature for distillation was set to 2. The memory buffer was allowed to store 100 examples per task for the replay methods. DGR utilized a multilayer perceptron with two layers as its generator.

We assessed the performance using the pixel-based metrics of intersection over union (IoU), $F1$-score, OA, and Kappa. The IoU metric evaluates the pixel-wise consistency between the predicted and the true building footprint extraction results. It can be defined mathematically as follows:

$$\text{IoU} = \frac{1}{k+1}\sum_{i=0}^{k}\frac{\text{TP}_i}{\text{FN}_i + \text{FP}_i + \text{TP}_i} \tag{15}$$

where k represents the number of classes (i.e., $k = 1$). TP_i represents the number of true positives of the ith class, while FP_i and FN_i represent the number of false positives and false negatives, respectively.

C. Experimental Results for the Global-CL Dataset

We presented the quantitative results in Table II to evaluate the continuous building footprint extraction performance. The proposed LAPNet framework achieves an IoU of 66.32% by leveraging global-scale building knowledge. However, its layout anchoring mechanism faces challenges in domain-incremental continual learning with heterogeneous data, resulting in an OA of 89.48%, which does not fully meet practical application requirements. The proposed LAPNet framework achieves a 10.59% IoU improvement over the naïve approach and surpasses unified and individual training frameworks. This highlights the layout anchoring mechanism's ability to select the most representative knowledge for dynamic updates toward global optimization. The proposed LAPNet framework performs well in the IoU metric, addressing knowledge forgetting by accurately distinguishing foreground and background across domains. The $F1$-score reflects its ability to balance knowledge accumulation, while the OA metric highlights its overall performance with layout features as a universal geospatial representation. The Kappa metric demonstrates LAPNet's effective retention of early-domain knowledge, ensuring reliable continuous building footprint extraction. Parameter regularization methods retain important information effectively but face challenges with large domain gaps, with IoUs of 57.07% (EWC) and 55.88% (SI). Functional regularization methods effectively transfer knowledge but tend to forget early-domain memories. BiC and LGKD achieve impressive IoU scores of 64.89% and 64.37%, respectively. Replay methods, such as ER, perform well in complex scenarios but neglect geospatial knowledge, hindering the selection of representative experiences. LAPNet outperforms existing continual semantic segmentation methods in handling foreground–background confusion, with a 1.95% IoU improvement over LGKD, demonstrating the effectiveness of its layout anchoring strategy for stable optimization in complex remote sensing scenarios.

TABLE II
COMPARATIVE EXPERIMENTAL RESULTS WERE OBTAINED USING STATE-OF-THE-ART DOMAIN-INCREMENTAL CONTINUAL LEARNING METHODS ON THE GLOBAL-CL DATASET. THE BEST RESULTS ARE IN RED AND THE SECOND BEST RESULTS ARE IN BLUE

Methods		IoU (%)	F1 (%)	OA (%)	Kappa
Naïve		55.73	65.25	84.75	0.33
Unified training		64.40	75.06	87.65	0.50
Individual training		64.44	74.72	88.83	0.50
Parameter regularization	EWC [19]	57.07	66.42	86.19	0.34
Parameter regularization	SI [20]	55.88	65.17	85.55	0.32
Functional regularization	LwF [23]	56.78	66.23	85.92	0.34
Functional regularization	ILT [46]	51.49	63.66	78.85	0.30
Functional regularization	BiC [47]	64.89	75.39	88.10	0.51
Functional regularization	MiB [48]	56.46	67.71	82.36	0.36
Functional regularization	PLOP [49]	63.36	72.81	85.24	0.44
Functional regularization	LGKD [50]	64.37	75.71	87.18	0.45
Replay	ER [51]	65.90	76.11	89.45	0.52
Replay	iCaRL [31]	65.84	75.94	89.44	0.52
Replay	GSS-Greedy [32]	65.52	75.96	88.76	0.52
Replay	A-GEM [26]	60.38	70.38	87.47	0.42
Replay	DGR [27]	54.91	63.84	85.64	0.30
Latent layout anchoring (proposed)		66.24	76.40	89.43	0.53
Layout-aware prioritizing learning (proposed)		65.98	76.13	89.44	0.53
LAPNet (proposed)		66.32	76.47	89.48	0.55

Fig. 6 displays the IoU variation graphs comparing LAPNet and the naïve approach for continuous building footprint extraction. Results indicate the IoU improvements when comparing the proposed LAPNet framework with the naïve approach across six tasks are 10.44%, 5.19%, 3.39%, 22.30%, 20.87%, and 1.35%, respectively. The results indicate that the proposed LAPNet framework continually accumulates knowledge as the task sequence expands, ensuring a consistent improvement in accuracy. However, the naïve approach experiences accuracy declines for Task 1→2 and Task 5→6, revealing knowledge-forgetting issues during cross-task transitions. Overall, the proposed LAPNet framework exhibits a stable domain-incremental continual learning performance, while the naïve approach shows a significant variance in accuracy, demonstrating that the proposed LAPNet framework effectively deals with knowledge-forgetting issues.

Large-scale visualization results demonstrate that the proposed LAPNet framework maintains robust accuracy across regions and sensors on a global scale (see Fig. 7). In cross-regional tasks (Tasks 1, 4, and 5), the proposed LAPNet framework accurately identifies the uniformly arranged buildings in Europe and America (Task 1), effectively extracts the dense buildings in Asia (Task 4), and precisely recalls the sparse buildings in Africa while efficiently eliminating background elements such as hills. In cross-sensor tasks (Tasks 2, 3, and 6), the proposed LAPNet framework conducts urbanization analysis using the GF-2 satellite sensor in different regions (Hongshan and Jiang'an) and during different periods (2015, 2016, and 2018). From 2015 to 2018, Hongshan experienced rapid development with a significant

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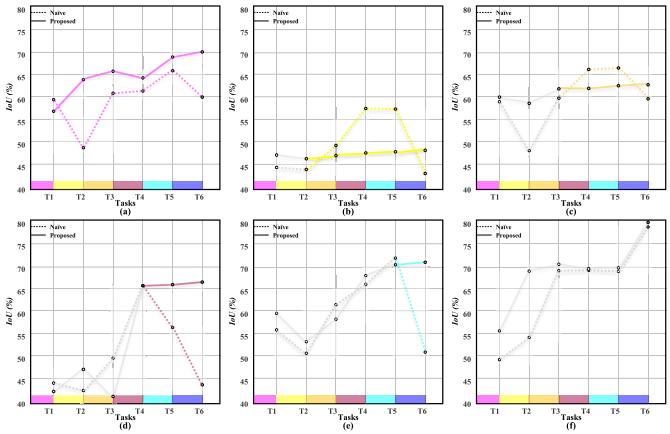


Fig. 6. IoU variation graphs comparing the proposed LAPNet framework (solid lines) with the naïve approach (dashed lines) across different stages of continuous learning for tasks 1–6, labeled (a)–(f). Gray lines indicate evaluation without learning on the current task, whereas colored lines indicate evaluation after learning on the current task.

increase in building density, whereas Jiang'an, already economically developed, showed minimal urbanization changes during the same period. The proposed LAPNet framework exhibits inferior performance in Jiang'an, primarily due to its layout anchoring mechanism. While this mechanism clusters representative experiences using geospatial layout knowledge, it limits the framework's adaptability to heterogeneous data from different satellite sensors, where significant radiometric differences further degrade its performance. UAV-borne sensor images capture the detailed building patterns in Nanjing (Task 3) and Tallinn (Task 6), allowing the proposed LAPNet framework to generate regular building boundaries.

Fig. 8 indicates that the proposed LAPNet framework can efficiently recall building footprints and identify diverse building patterns in cross-sensor and cross-region tasks, while other approaches are impaired by the challenge of spatial layout heterogeneity. LAPNet captures fine-grained features to ensure high recall by distinguishing foreground from background, whereas other approaches prioritize classifying pixels as background due to the larger proportion of background. The discrepancies across sensors result in radiation variations, with satellite sensors being more susceptible to atmospheric effects on radiation transmission than UAV-borne sensors, leading to diverse spectral responses. In cross-sensor tasks (Tasks 2, 3, and 6), the results of Task 2 indicate that the GF-2 sensor severely impairs building footprint extraction due to radiation variations, leading to missed detection of building objects. On the other hand, the differences across regions result in diverse terrains and landscapes. For example, African regions are dominated by hills and vegetation, while European and American areas are characterized by plains and impervious surfaces, leading to variations in materials and layouts. In cross-region tasks (Tasks 1, 4, and 5), recalling building footprints with varying materials and layouts presents a challenge, as building objects are often misclassified as background. Although knowledge forgetting is more prominent in the early stages, such as Task 1, the LAPNet framework excels in accurately extracting building footprints and determining precise boundary locations.

TABLE III

QUANTITATIVE RESULTS FOR THE PROPOSED LATENT LAYOUT ANCHORING MODULE AND OTHER STORAGE STRATEGIES

Storage strategy	IoU (%)
Random	65.90
Closet	65.80
Herding [37]	65.87
Grad-closet [32]	65.59
Grad-IQP [32]	65.52
Our module (foreground)	65.63
Our module (background)	**66.24**

IV. DISCUSSION

A. Ablation Experiments for the Latent Layout Anchoring

Table III summarizes the quantitative results for the proposed latent layout anchoring module and other storage strategies. Random storage serves as the baseline. The closet and herding strategies use semantic features to cluster experiences, while the Grad-closet and Grad-IQP strategies use gradient vectors to cluster experiences. The proposed latent layout anchoring module obtains the best performance with an IoU of 66.24%. Other storage strategies are impaired by the spatial layout heterogeneity problem since they aggregate global information for clustering, leading to convergence on local optima that may not contain the most representative experiences. To demonstrate the performance of the layout prior, we applied the layout prior indexing mechanism $\mathcal{P}$ to the foreground and background, respectively. The results indicate that utilizing the layout prior with background features yields better accuracy. This is attributed to the fact that spatial layout heterogeneity primarily arises from complex backgrounds. Clustering representative experiences based on background features eliminates irrelevant information, thus enhancing the ability to retain essential knowledge.

Fig. 9 presents an image-level clustering visualization of the foreground and background of experiences stored in the memory buffer. Each data point in the plot represents either the foreground or background corresponding to a specific experience. The t-SNE was applied for dimensionality reduction, with principal component analysis as the initial step, followed by t-SNE optimization with an early exaggeration parameter set to 30 to ensure proper separation of data points across categories. The results demonstrate that the proposed latent layout anchoring module effectively reduces background noise by dynamically storing layout-anchored experiences. This module produces more compact clusters, suggesting fewer noisy perturbations are introduced into the memory buffer. Additionally, the module efficiently identifies and clusters complex backgrounds, leading to better aggregating background information around the clusters and generating more representative experiences.

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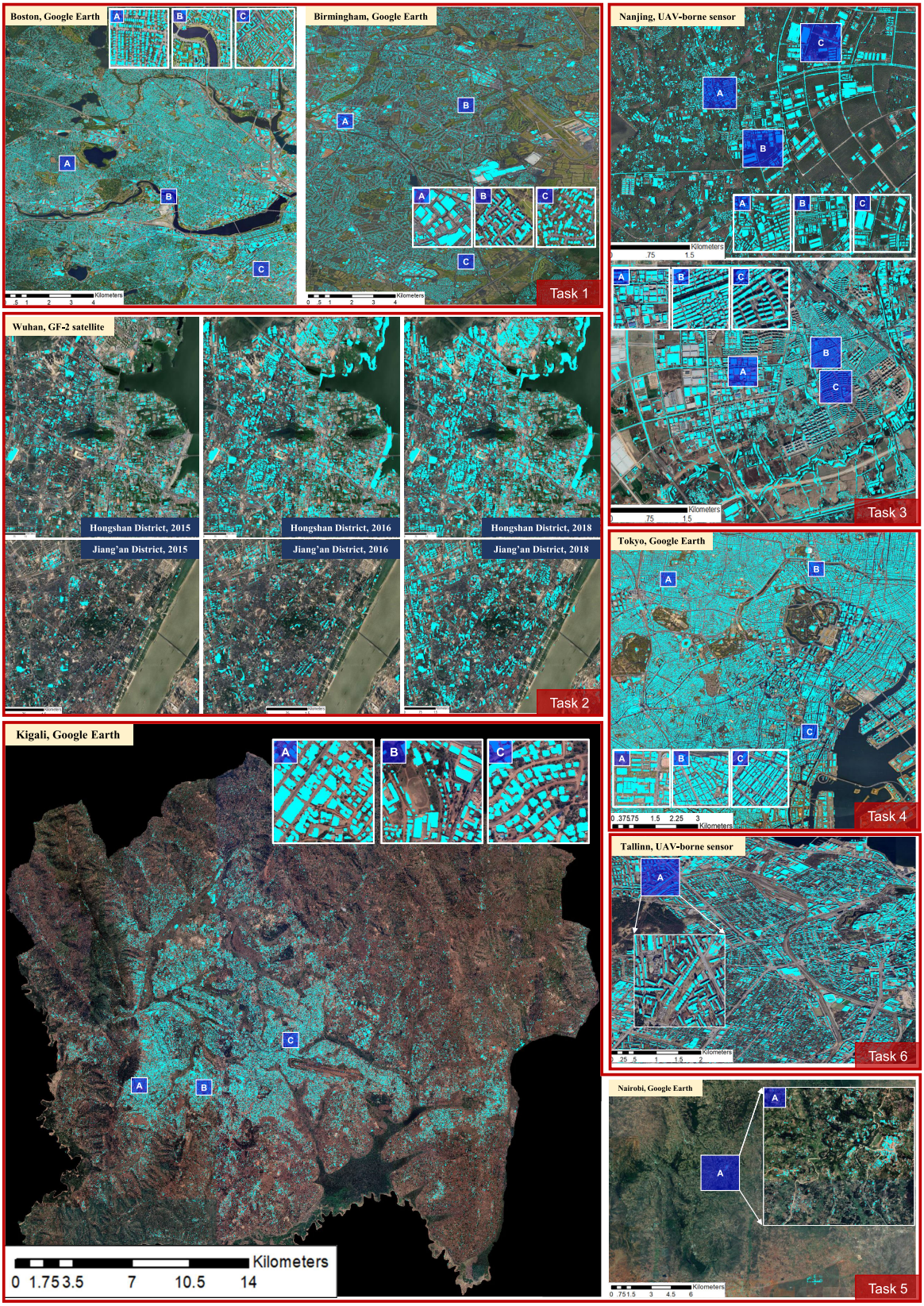


Fig. 7. Large-scale visualization results of the Global-CL dataset, including GE images from Europe (Birmingham) for Task 1, America (Boston) for Task 2, GF-2 data collected in Wuhan for Task 3, UAV-borne sensor images from Nanjing for Task 4, GE images from Asia (Tokyo) for Task 5, and GE images from Africa (Nairobi and Kigali) as well as UAV-borne sensor images from Tallinn for Task 6. A–C represent zoomed-in views.

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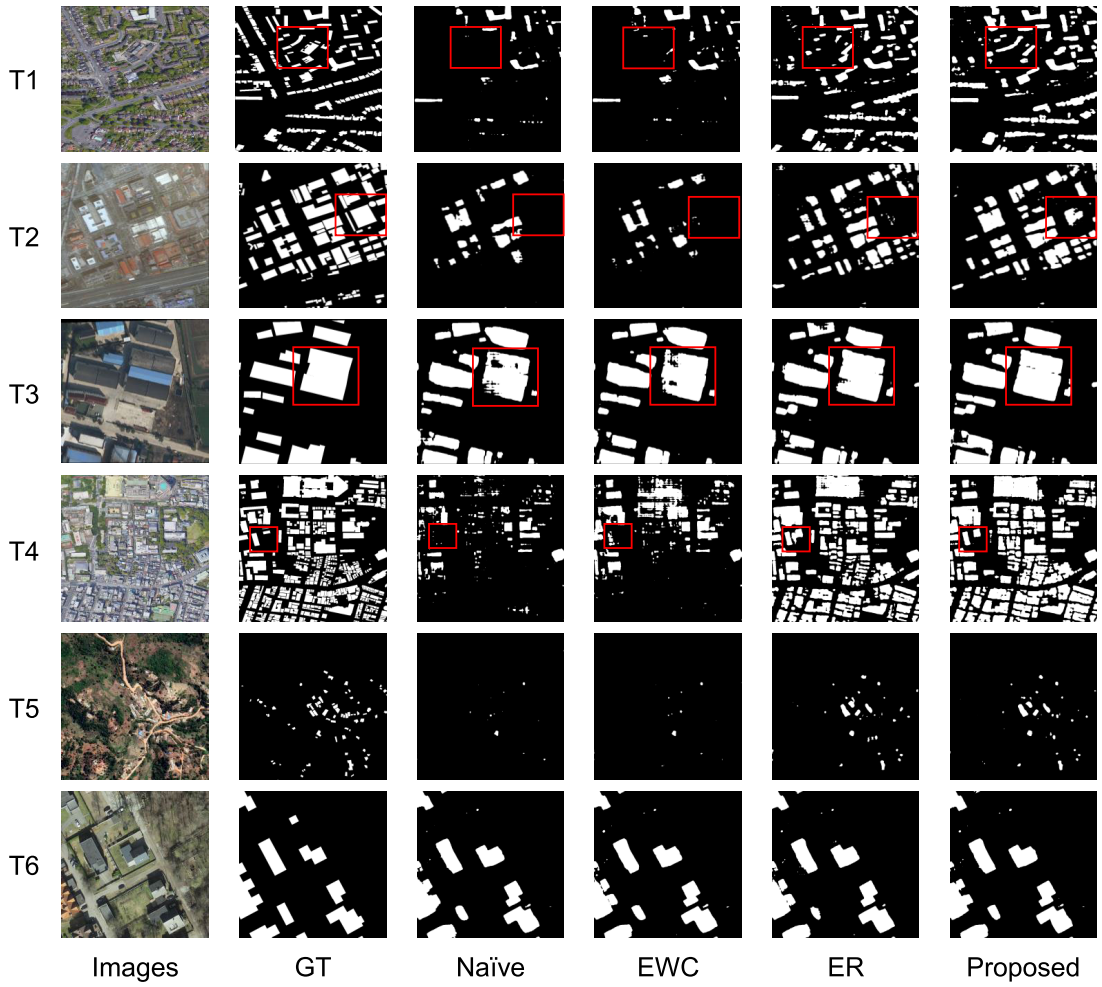


Fig. 8. Visualization results of continuous building footprint extraction in cross-sensor and cross-region tasks. Task 1: European and American regions, Task 2: Wuhan (GF-2 sensor), Task 3: Nanjing (UAV-borne sensor), Task 4: Asian regions, Task 5: African regions, and Task 6: Tallinn (UAV-borne sensor).

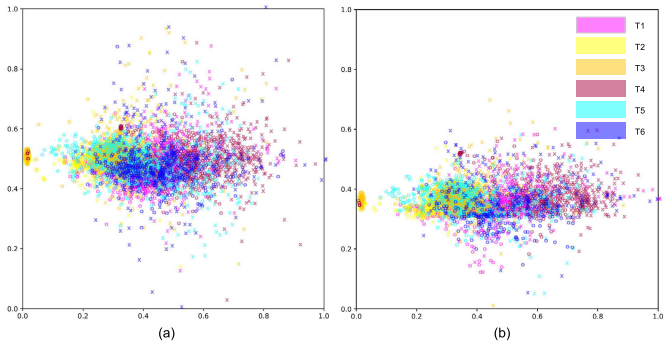


Fig. 9. Image-level clustering visualization of experiences stored in the memory buffer. (a) Random storage and (b) latent layout anchoring. "o:" foreground, "x:" background. T1: European and American regions, T2: Wuhan (GF-2 sensor), T3: Nanjing (UAV-borne sensor), T4: Asian regions, T5: African regions, and T6: Tallinn (UAV-borne sensor).

TABLE IV

ABLATION RESULTS ON THE KERNEL SIZE (k) OF THE SPATIAL LAYOUT ATTENTION F_{att} IN THE LAYOUT-AWARE PRIORITIZING LEARNING MODULE

Kernel size k	IoU (%)
3	**65.98**
5	65.92
7	65.86

B. Ablation Experiments for the Layout-Aware Prioritizing Learning

As exhibited in Table IV, we conducted experiments on the convolutional kernel size (k) of the spatial layout attention F_{att} to analyze the impact of the spatial receptive field on capturing discriminative layout features. The experimental results reveal that the optimal performance (IoU = 65.98%) is achieved when $k = 3$. This suggests that using a small receptive field to capture layout features allows for fine-grained detail information extraction, contributing to the effective aggregation of critical features for layout-aware prioritized learning.

To analyze the effectiveness of prioritized learning, we summarized the results of ablation experiments in Table V. The proposed layout-aware prioritized learning module incorporates spatial layout attention and prioritized learning to achieve superior performance. Notably, prioritized learning demonstrates a 0.21% improvement in IoU compared to spatial layout attention alone. This modest enhancement can be attributed to prioritized learning's ability to identify the most critical

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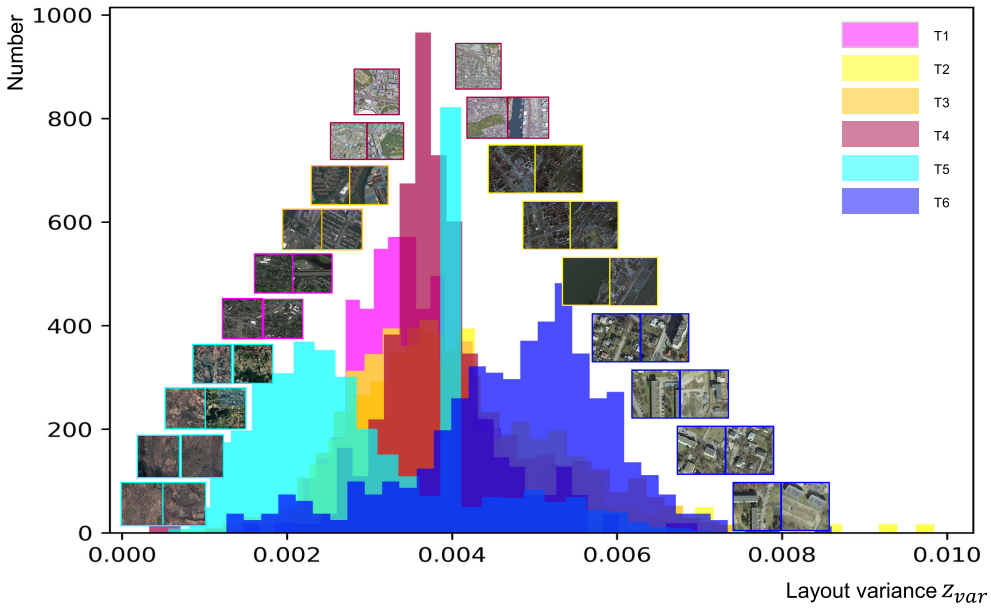


Fig. 10. Probability density distribution of the layout variance metric Z_{var} in the layout-aware prioritizing learning module ranked from low to high is as follows: African regions (Task 5), European and American regions (Task 1), Nanjing (UAV-borne sensor, Task 3), Asian regions (Task 4), Wuhan (GF-2 sensor, Task 2), and Tallinn (UAV-borne sensor, Task 6).

TABLE V

ABLATION RESULTS OF THE PROPOSED LAYOUT-AWARE PRIORITIZED LEARNING MODULE, INCLUDING SPATIAL LAYOUT ATTENTION AND PRIORITIZED LEARNING

Spatial layout attention	Prioritizing learning	IoU (%)
		65.90
✓		65.77
✓	✓	**65.98**

knowledge dynamically, thus maximizing the interclass distance and improving model training efficiency.

Fig. 10 illustrates the probability density distribution of the layout variance metric Z_{var}. Results show that the layout variance metric indicates spatial layout heterogeneity, highlighting the diversity of sensor observations and the complexity of geospatial layouts. In cross-sensor tasks (Tasks 2, 3, and 6), the layout variance metric highlights the variability in sensor observations. Specifically, satellite sensors exhibit lower layout variance than UAV-borne sensors. Satellite sensors have stable observation altitudes and orientations, resulting in a relatively homogeneous spatial layout. Nonetheless, we observe a diverse range of layout variances in the case of UAV images with a resolution of 0.09 m (Task 6). This is primarily due to the substantial randomness in local details depicted by ultrahigh-resolution UAV images. In cross-regional tasks (Tasks 1, 4, and 5), the layout variance metric indicates the complexity of geospatial layouts. Specifically, African regions exhibit lower layout variance than European and American regions. This is because European and American regions (Task 1) are dominated by well-established man-made infrastructure, resulting in complex spatial layouts. A low layout variance is observed in the African building pattern (Task 5). African buildings exhibit weak foreground signals and present ambiguous foreground–background information.

C. Global-Scale Analysis of the Spatial Layout Heterogeneity

We conducted experiments with different task sequences to demonstrate the impact of spatial layout heterogeneity on continuous building footprint extraction. The tasks were ordered based on their test set accuracy.

The results in Table VI reveal that arranging the tasks in ascending order leads to the best performance, with an IoU of 66.32%. This is because high-accuracy tasks represent greater model training efficiency, and introducing high-accuracy tasks later in the sequence can progressively guide the model to acquire discriminative knowledge, thereby establishing knowledge connections across diverse tasks. Swapping the sequences of cross-sensor and cross-region tasks has minimal impact on accuracy improvement. This is because, although spatial layout heterogeneity is divided into cross-sensor and cross-region settings, their fundamental difference lies in the layout disparity between foreground and background. Fig. 11 provides a detailed illustration of the performance of task sequences in terms of accuracy improvement. In Fig. 11(a), it is evident that the late sequence (−18.26 or 13.50) outperforms the early sequence (−18.34 or 12.46). Fig. 11(b) demonstrates that swapping the sequences of cross-sensor and cross-region has no significant impact.

Fig. 12 displays the t-SNE visualization results of encoder features, providing insights into the spatial layout heterogeneity among various tasks. These visualizations depict the distinct separation of tasks within the feature space, indicating significant spatial layout heterogeneity in continuous building footprint extraction. On the other hand, shared semantic features exist between tasks, facilitating effective knowledge correlation across different tasks and enhancing cross-task accuracy during knowledge accumulation. It is important to note that although the proposed LAPNet framework requires a straightforward hyperplane to distinguish foreground from

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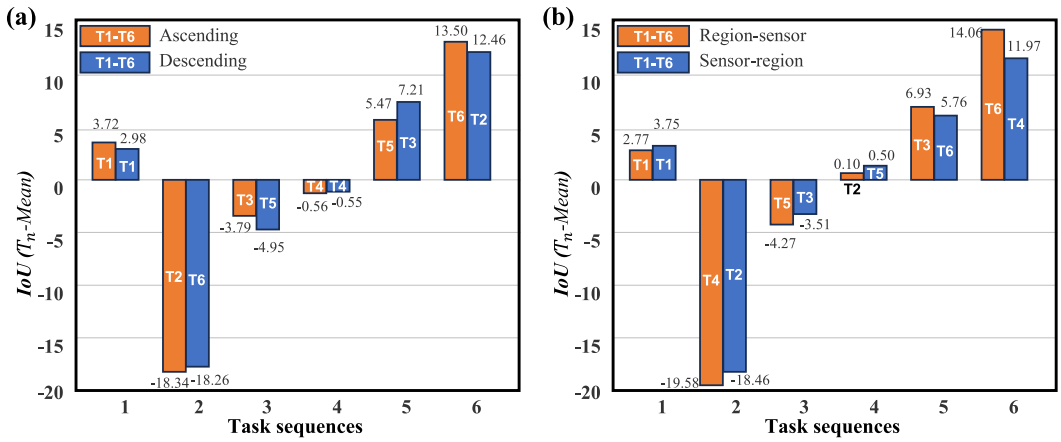


Fig. 11. Ablation analysis on the task sequences of continuous building footprint extraction. (a) "Ascending" and "Descending" refer to sorting based on the accuracy of independent training strategies. (b) "Region-sensor" and "Sensor-region" refer to sorting first by cross-region and then by cross-sensor tasks (or vice versa). T1: European and American regions, T2: Wuhan (GF-2 sensor), T3: Nanjing (UAV-borne sensor), T4: Asian regions, T5: African regions, and T6: Tallinn (UAV-borne sensor).

TABLE VI

ABLATION RESULTS ON TASK SORTING. "ASCENDING" AND "DESCENDING" REFER TO SORTING BASED ON THE ACCURACY OF INDEPENDENT TRAINING STRATEGIES, WHEREAS "REGION-SENSOR" AND "SENSOR-REGION" REFER TO SORTING FIRST BY CROSS-REGION AND THEN BY CROSS-SENSOR TASKS (OR VICE VERSA)

Sorting	Task Id	IoU (%)	F1 (%)	OA (%)	Kappa
Ascending	**123456**	**66.32**	**76.47**	**89.48**	**0.55**
Descending	165432	64.99	75.75	87.45	0.53
Region-sensor	145236	65.44	75.85	88.12	0.54
Sensor-region	123645	65.24	75.91	87.84	0.54

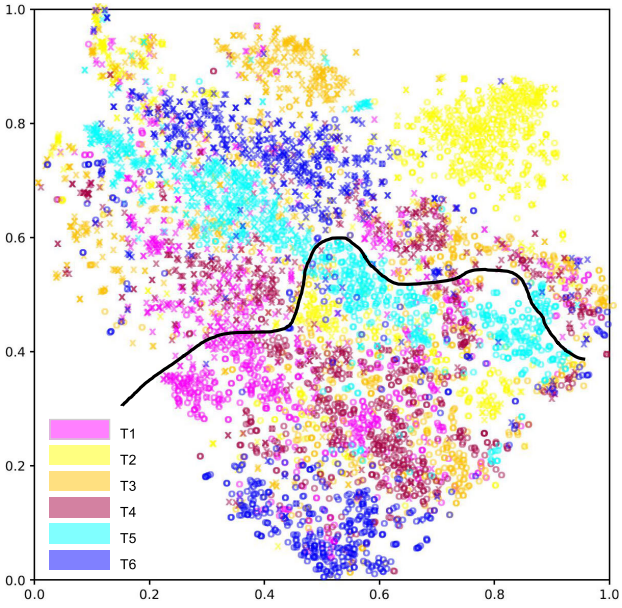


Fig. 12. t-SNE visualization results of the proposed LAPNet framework's encoder features. This reveals the spatial layout heterogeneity among different tasks. T1 for European and American regions, T2 for Wuhan (GF-2 sensor), T3 for Nanjing (UAV-borne sensor), T4 for Asian regions, T5 for African regions, and T6 for Tallinn (UAV-borne sensor). "o:" foreground, "x:" background.

background, a misclassification challenge still exists due to semantic confusion arising from spatial layout heterogeneity.

TABLE VII

COMPARATIVE EXPERIMENTS ON PSEUDO AND GROUND TRUTH LABELS BASED ON THE RANDOMLY SELECTED 600 SAMPLES

Methods	Labels	IoU (%)	F1 (%)	OA (%)	Kappa
Naïve	Synthetic	52.04	63.91	77.37	0.33
	Real	53.31	64.89	78.58	0.34
ER	Synthetic	61.39	73.20	84.50	0.48
	Real	62.52	74.30	80.16	**0.53**
LAPNet (proposed)	Synthetic	61.84	73.57	85.23	0.48
	Real	**64.17**	**75.36**	**87.46**	0.51

D. Comparing Pseudo and Ground Truth Labels

To evaluate label quality, we randomly selected 100 samples from each task (Tasks 1–6) to construct a training set with pseudo labels. Tasks 2–6 were manually annotated to build a training set with ground truth labels. Experiments were conducted to compare the naïve approach, the ER approach, and the proposed LAPNet framework, where the memory buffer stored ten examples per task for both the ER approach and the LAPNet framework.

Table VII shows that ground truth labels consistently improve accuracy across all comparison methods compared to pseudo labels, highlighting that precise supervision promotes stable model optimization during knowledge accumulation. The proposed LAPNet framework achieved the most significant IoU increase of 2.33%, demonstrating its ability to leverage the high-precision information from ground truth labels to uncover latent domain relationships. Furthermore, the LAPNet framework showed IoU improvements of 0.45% and 1.65% over the advanced ER approach when using pseudo and ground truth labels, respectively. This indicates that the proposed LAPNet framework effectively achieves continuous building footprint extraction by utilizing layout-anchored prioritizing learning.

V. CONCLUSION

In this article, we have proposed a domain-incremental continual learning algorithm named LAPNet designed for

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continuous building footprint extraction. To solve the issue of knowledge forgetting due to spatial layout heterogeneity, the LAPNet framework consists of a latent layout anchoring module to narrow the background diversity and a layout-aware prioritizing learning module to maximize the foreground–background distance. Experiments conducted on the Global-CL dataset illustrate that the proposed LAPNet framework achieves a 10.59% IoU improvement over the naïve approach and a 0.42% increase over the state-of-the-art ER approach. The experiments showed that the proposed LAPNet framework effectively handles radiation variations caused by cross-sensors (an IoU of 47.98% for GF-2 sensors) and terrain and landscape divergencies caused by cross-regions (an IoU of 71.79% in African regions). The proposed LAPNet framework effectively establishes knowledge connections across diverse domains and significantly enhances global building footprint extraction performance. In future work, we aim to develop a dynamic classifier that adapts to radiometric differences across satellite sensors, enabling effective handling of multisource data.

REFERENCES

[1] X. Liu et al., "High-spatiotemporal-resolution mapping of global urban change from 1985 to 2015," *Nature Sustainability*, vol. 3, no. 7, pp. 564–570, May 2020.

[2] T. Wiedmann and C. Allen, "City footprints and SDGs provide untapped potential for assessing city sustainability," *Nature Commun.*, vol. 12, no. 1, p. 3758, Jun. 2021.

[3] Y. Liu, D. Chen, A. Ma, Y. Zhong, F. Fang, and K. Xu, "Multiscale U-shaped CNN building instance extraction framework with edge constraint for high-spatial-resolution remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 6106–6120, Jul. 2021.

[4] J. Yuan, "Learning building extraction in aerial scenes with convolutional networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 11, pp. 2793–2798, Nov. 2018.

[5] J. Wang, L. Meng, W. Li, W. Yang, L. Yu, and G.-S. Xia, "Learning to extract building footprints from off-nadir aerial images," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 1, pp. 1294–1301, Jan. 2023.

[6] Y. Ju, I. Dronova, and X. Delclòs-Alió, "A 10 m resolution urban green space map for major Latin American cities from Sentinel-2 remote sensing images and OpenStreetMap," *Sci. Data*, vol. 9, no. 1, p. 586, Sep. 2022.

[7] M. Pesaresi et al., "A global human settlement layer from optical HR/VHR RS data: Concept and first results," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 6, no. 5, pp. 2102–2131, Oct. 2013.

[8] J. Martínez, G. Mboup, R. Sliuzas, and A. Stein, "Trends in urban and slum indicators across developing world cities, 1990–2003," *Habitat Int.*, vol. 32, no. 1, pp. 86–108, Mar. 2008.

[9] J. Gao and B. C. O'Neill, "Mapping global urban land for the 21st century with data-driven simulations and shared socioeconomic pathways," *Nature Commun.*, vol. 11, no. 1, p. 2302, May 2020.

[10] I. McCallum et al., "Estimating global economic well-being with unlit settlements," *Nature Commun.*, vol. 13, no. 1, p. 2459, May 2022.

[11] K. C. Seto, M. Fragkias, B. Güneralp, and M. K. Reilly, "A meta-analysis of global urban land expansion," *PLoS ONE*, vol. 6, no. 8, Aug. 2011, Art. no. e23777.

[12] D. Chen, Y. Zhong, Z. Zheng, A. Ma, and X. Lu, "Urban road mapping based on an end-to-end road vectorization mapping network framework," *ISPRS J. Photogramm. Remote Sens.*, vol. 178, pp. 345–365, Aug. 2021.

[13] D. Chen, A. Ma, Z. Zheng, and Y. Zhong, "Large-scale agricultural greenhouse extraction for remote sensing imagery based on layout attention network: A case study of China," *ISPRS J. Photogramm. Remote Sens.*, vol. 200, pp. 73–88, Jun. 2023.

[14] Z. Zheng, Y. Zhong, J. Wang, A. Ma, and L. Zhang, "Building damage assessment for rapid disaster response with a deep object-based semantic change detection framework: From natural disasters to man-made disasters," *Remote Sens. Environ.*, vol. 265, Nov. 2021, Art. no. 112636.

[15] M. Delange et al., "A continual learning survey: Defying forgetting in classification tasks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 7, pp. 3366–3385, Jul. 2021.

[16] D. Kiyasseh, T. Zhu, and D. Clifton, "A clinical deep learning framework for continually learning from cardiac signals across diseases, time, modalities, and institutions," *Nature Commun.*, vol. 12, no. 1, p. 4221, Jul. 2021.

[17] D. Ye, J. Peng, H. Li, and L. Bruzzone, "Better memorization, better recall: A lifelong learning framework for remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5626814.

[18] J. Ge, H. Tang, N. Yang, and Y. Hu, "Rapid identification of damaged buildings using incremental learning with transferred data from historical natural disaster cases," *ISPRS J. Photogramm. Remote Sens.*, vol. 195, pp. 105–128, Jan. 2023.

[19] K. James et al., "Overcoming catastrophic forgetting in neural networks," *Proc. Nat. Acad. Sci. USA*, vol. 114, no. 13, pp. 3521–3526, Mar. 2017.

[20] F. Zenke, B. Poole, and S. Ganguli, "Continual learning through synaptic intelligence," in *Proc. Int. Conf. Mach. Learn.*, 2017, pp. 3987–3995.

[21] X. Rui, Z. Li, Y. Cao, Z. Li, and W. Song, "DILRS: Domain-incremental learning for semantic segmentation in multi-source remote sensing data," *Remote Sens.*, vol. 15, no. 10, p. 2541, May 2023.

[22] M. A. Ma'Sum, M. Pratama, R. Savitha, L. Liu, Habibullah, and R. Kowalczyk, "Unsupervised few-shot continual learning for remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4707214.

[23] Z. Li and D. Hoiem, "Learning without forgetting," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 12, pp. 2935–2947, Dec. 2017.

[24] K. Shen, X. Yang, S. Lolli, and G. Vivone, "A continual learning-guided training framework for pansharpening," *ISPRS J. Photogramm. Remote Sens.*, vol. 196, pp. 45–57, Feb. 2023.

[25] O. Tasar, Y. Tarabalka, and P. Alliez, "Incremental learning for semantic segmentation of large-scale remote sensing data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 9, pp. 3524–3537, Sep. 2019.

[26] A. Chaudhry, M. Ranzato, M. Rohrbach, and M. Elhoseiny, "Efficient lifelong learning with A-GEM," 2018, *arXiv:1812.00420*.

[27] H. Shin, J. K. Lee, J. Kim, and J. Kim, "Continual learning with deep generative replay," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 2994–3003.

[28] J. Wang, S. Guo, Z. Hua, R. Huang, J. Hu, and M. Gong, "CL-CaGAN: Capsule differential adversarial continual learning for cross-domain hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5517315.

[29] W. Dong, Y. Chen, S. Xiao, J. Qu, and Y. Li, "Incremental detection of hyperspectral targets in consistent scenes with continuous learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5531513.

[30] R. Tiwari, K. Killamsetty, R. Iyer, and P. Shenoy, "GCR: Gradient coreset based replay buffer selection for continual learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 99–108.

[31] S.-A. Rebuffi, A. Kolesnikov, G. Sperl, and C. H. Lampert, "ICaRL: Incremental classifier and representation learning," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2001–2010.

[32] R. Aljundi, M. Lin, B. Goujaud, and Y. Bengio, "Gradient based sample selection for online continual learning," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 32, 2019, pp. 11817–11826.

[33] M. Perkonigg et al., "Dynamic memory to alleviate catastrophic forgetting in continual learning with medical imaging," *Nature Commun.*, vol. 12, no. 1, p. 5678, Sep. 2021.

[34] C. D. Kim, J. Jeong, and G. Kim, "Imbalanced continual learning with partitioning reservoir sampling," in *Proc. 16th Eur. Conf. Comput. Vision (ECCV)*, Glasgow, U.K., Jan. 2020, pp. 411–428.

[35] K.-H. Li, "Reservoir-sampling algorithms of time complexity $O(n(1+\log(N/n)))$," *ACM Trans. Math. Softw.*, vol. 20, no. 4, pp. 481–493, Dec. 1994.

[36] T. Schaul, J. Quan, I. Antonoglou, and D. Silver, "Prioritized experience replay," 2015, *arXiv:1511.05952*.

[37] L.-C. Chen, G. Papandreou, F. Schroff, and H. Adam, "Rethinking atrous convolution for semantic image segmentation," 2017, *arXiv:1706.05587*.

[38] M. Lin, Q. Chen, and S. Yan, "Network in network," 2013, *arXiv:1312.4400*.

[39] M. Welling, "Herding dynamical weights to learn," in *Proc. 26th Annu. Int. Conf. Mach. Learn.*, Jun. 2009, pp. 1121–1128.

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[40] C. M. Bishop and N. M. Nasrabadi, *Pattern Recognition and Machine Learning*, vol. 4. Cham, Switzerland: Springer, 2006.
[41] F. Milletari, N. Navab, and S. Ahmadi, "V-Net: Fully convolutional neural networks for volumetric medical image segmentation," in *Proc. 4th Int. Conf. 3D Vis. (3DV)*, Oct. 2016, pp. 565–571.
[42] D. Chen, A. Ma, and Y. Zhong, "Semi-supervised knowledge distillation framework for global-scale urban man-made object remote sensing mapping," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 122, Aug. 2023, Art. no. 103439.
[43] A. Paszke et al., "PyTorch: An imperative style, high-performance deep learning library," in *Proc. 33rd Conf. Neural Inf. Process. Syst.*, 2019, pp. 8026–8037.
[44] Z. Cheng, H. Sun, M. Takeuchi, and J. Katto, "Deep residual learning for image compression," 2019, *arXiv:1906.09731*.
[45] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A large-scale hierarchical image database," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, May 2009, pp. 248–255.
[46] U. Michieli and P. Zanuttigh, "Incremental learning techniques for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshop (ICCVW)*, Oct. 2019, pp. 3205–3212.
[47] Y. Wu et al., "Large scale incremental learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 374–382.
[48] F. Cermelli, M. Mancini, S. R. Bulò, E. Ricci, and B. Caputo, "Modeling the background for incremental learning in semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9233–9242.
[49] A. Douillard, Y. Chen, A. Dapogny, and M. Cord, "PLOP: Learning without forgetting for continual semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Nashville, TN, USA, Jun. 2021, pp. 4040–4050.
[50] Z. Yang et al., "Label-guided knowledge distillation for continual semantic segmentation on 2D images and 3D point clouds," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 18555–18566.
[51] P. Buzzega, M. Boschini, A. Porrello, and S. Calderara, "Rethinking experience replay: A bag of tricks for continual learning," in *Proc. 25th Int. Conf. Pattern Recognit. (ICPR)*, Jan. 2021, pp. 2180–2187.



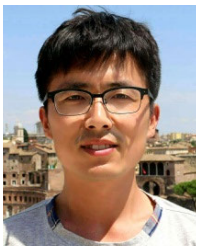
Dingyuan Chen received the B.S. degree from the School of Geography and Information Engineering, China University of Geosciences, Wuhan, China, in 2019, and the M.E. and Ph.D. degrees from Wuhan University, Wuhan, in 2021 and 2024, respectively.

He is currently an Associate Professor with the Space Information Research Institute, Hangzhou Dianzi University, Hangzhou, China. His research interests include building extraction with multisource remote sensing imagery and deep learning-based object detection.



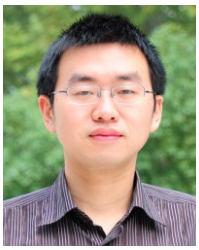
Zhaohui Song (Senior Member, IEEE) received the B.S., M.E., and Ph.D. degrees from Harbin Institute of Technology, Harbin, China, in 1992, 1998, and 2006, respectively.

He was a Professor at Harbin Institute of Technology. He has been a Professor with Hangzhou Dianzi University, Hangzhou, China, since 2022. His research interests include antenna theory and technology, microwave circuits and systems, electromagnetic wave propagation theory, space information perception, and transmission.



Ailong Ma (Senior Member, IEEE) received the B.S. degree from China University of Petroleum, Qingdao, China, in 2010, and the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2017.

He is now working as Research Associate with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing (LIESMARS), Wuhan University. He has published more than 15 research journals articles on remote sensing image processing. His major research interests include remote sensing image processing, pattern recognition, and deep learning.



Yanfei Zhong (Senior Member, IEEE) received the B.S. degree in information engineering and the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2002 and 2007, respectively.

Since 2010, he has been a Full Professor with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing (LIESMARS), Wuhan University. He organized the Intelligent Data Extraction, Analysis and Applications of Remote Sensing (RSIDEA) Research Group. He has published more than 100 research articles in international journals, such as *Remote Sensing of Environment*, *ISPRS Journal of Photogrammetry and Remote Sensing*, and IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING. His research interests include hyperspectral remote sensing information processing, high-resolution remote sensing image understanding, and geoscience interpretation for multisource remote sensing data and applications.

Dr. Zhong is a fellow of the Institution of Engineering and Technology (IET). He was a recipient of the 2016 Best Paper Theoretical Innovation Award from the International Society for Optics and Photonics (SPIE). He won the Second-Place Prize in 2013 IEEE GRSS Data Fusion Contest and the Single-View Semantic 3-D Challenge of the 2019 IEEE GRSS Data Fusion Contest, respectively. He is currently serving as an Associate Editor for IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING, and the *International Journal of Remote Sensing*.

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